YAYASAN MEMAJUKAN ILMU DAN KEBUDAYAAN
UNIVERSITAS SIBER ASIA
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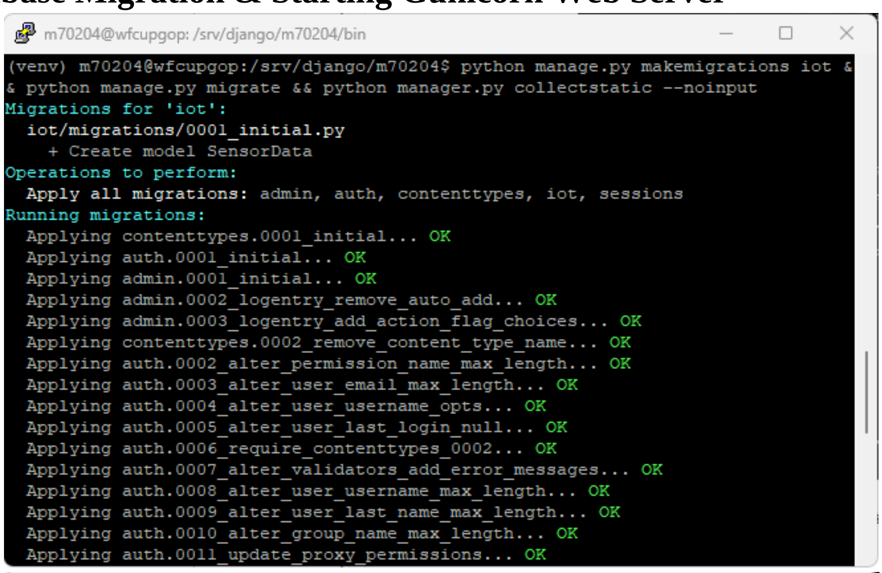
LEMBAR JAWABAN
ASSIGNMENT REPORT - WEEK 10 (TASK 1)
SEMESTER GENAP TAHUN AJARAN 2025/2026

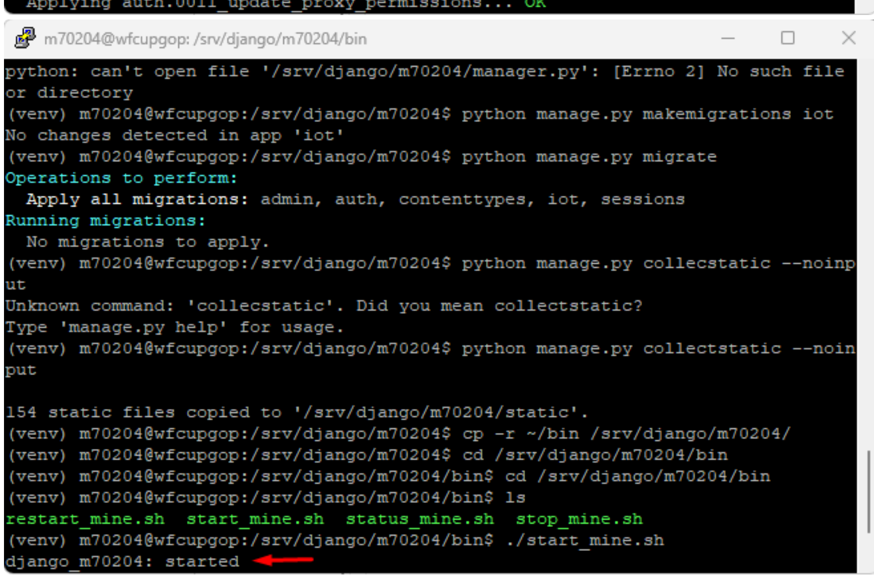
Mata Kuliah	: Internet of Things (IoT)
Kelas	: IF601
Prodi	: PJJ Informatika
Nama Mahasiswa	: Muhammad Ketsar Ali Abi Wahid
NIM	: 230401070204
Dosen	: Prof. Jong-Dae Park

<u>Jawaban</u>

Step 1: Database Migration & Starting Gunicorn Web Server



```
m70204@wfcupgop: /srv/django/m70204/bin
(venv) m70204@wfcupgop:/srv/django/m70204$ python manage.py makemigrations iot &
& python manage.py migrate && python manager.py collectstatic --noinput
Migrations for 'iot':
  iot/migrations/0001_initial.py
    + Create model SensorData
Operations to perform:
  Apply all migrations: admin, auth, contenttypes, iot, sessions
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
```



```
m70204@wfcupgop: /srv/django/m70204/bin
python: can't open file '/srv/django/m70204/manager.py': [Errno 2] No such file
or directory
(venv) m70204@wfcupgop:/srv/django/m70204$ python manage.py makemigrations iot
No changes detected in app 'iot'
(venv) m70204@wfcupgop:/srv/django/m70204$ python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, contenttypes, iot, sessions
Running migrations:
  No migrations to apply.
(venv) m70204@wfcupgop:/srv/django/m70204$ python manage.py collectstatic --noinp
ut
Unknown command: 'collectstatic'. Did you mean collectstatic?
Type 'manage.py help' for usage.
(venv) m70204@wfcupgop:/srv/django/m70204$ python manage.py collectstatic --noin
put

154 static files copied to '/srv/django/m70204/static'.
(venv) m70204@wfcupgop:/srv/django/m70204$ cp -r ~/bin /srv/django/m70204/
(venv) m70204@wfcupgop:/srv/django/m70204$ cd /srv/django/m70204/bin
(venv) m70204@wfcupgop:/srv/django/m70204/bin$ cd /srv/django/m70204/bin
(venv) m70204@wfcupgop:/srv/django/m70204/bin$ ls
restart_mine.sh  start_mine.sh  status_mine.sh  stop_mine.sh
(venv) m70204@wfcupgop:/srv/django/m70204/bin$ ./start_mine.sh
django_m70204: started
```

Connected to the server via PuTTY, activated the Python virtual environment (venv), and executed 'python manage.py migrate' to apply database migrations. Ran 'python manage.py collectstatic --noinput' to compile static assets. Finally, copied Gunicorn control scripts and successfully launched the WSGI web server using './start_mine.sh'.

Step 2: Local Simulation Client (send_django_mock_data.py)

send_django_mock_data.py × | dht11_django.ino | sensor_gauge.html 9+ | sensor_data.html PERTEMUAN 10\JAWABAN\iot\t

PERTEMUAN 10 > JAWABAN > send_django_mock_data.py

```python
import urllib.request
import json
import time
import random

url = "https://m70204.belajarhub.id/django/api/sensor/"
api_key = "abc123"
device_id = "device01"

print("=== Pengiriman Data Sensor Django Otomatis Dimulai ===")

for i in range(1, 11):
    # Data suhu dan kelembapan realistis
    temp = round(random.uniform(25.0, 31.5), 1)
    hum = round(random.uniform(55.0, 72.0), 1)

    payload = {
        "apikey": api_key,
        "device_id": device_id,
        "temperature": temp,
        "humidity": hum
    }

    # Encode ke format JSON
    json_data = json.dumps(payload).encode('utf-8')

    # Request POST
    req = urllib.request.Request(url, data=json_data, method='POST')
    req.add_header('Content-Type', 'application/json')

    try:
        with urllib.request.urlopen(req) as response:
            res_text = response.read().decode('utf-8')
            print(f"[{i}/10] Terkirim -> Suhu: {temp}°C, Kelembapan: {hum}%. Respon: {res_text}")
    except Exception as e:
        print(f"[{i}/10] Gagal mengirim data: {e}")

    time.sleep(1)

print("\n=== Pengiriman Data Selesai! ===")
print("Silakan cek halaman web Django Anda di:")
print("-> https://m70204.belajarhub.id/django/device/device01/")
print("-> https://m70204.belajarhub.id/django/device/device01/gauge/")
```

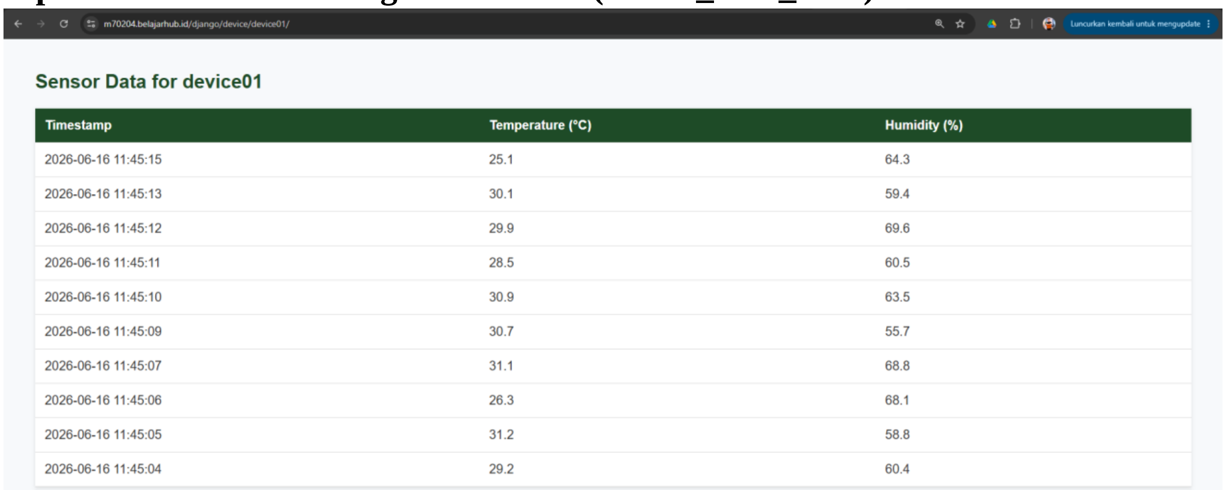
▷ Generate Mermaid Diagram | Open Chat @mermaid-chart

```
(base) PS E:\Documents\Documents Ketsar\UNSIA\MATKUL\SEMESTER 6\Internet of Things (IoT) - Prof. Jong-Dae Park\PERTEMUAN 10\JAWABAN> python .\send_django_mock_data.py
=== Pengiriman Data Sensor Django Otomatis Dimulai ===
[1/10] Terkirim -> Suhu: 29.2°C, Kelembapan: 60.4%. Respon: {"message":"Data saved"}
[2/10] Terkirim -> Suhu: 31.2°C, Kelembapan: 58.8%. Respon: {"message":"Data saved"}
[3/10] Terkirim -> Suhu: 26.3°C, Kelembapan: 68.1%. Respon: {"message":"Data saved"}
[4/10] Terkirim -> Suhu: 31.1°C, Kelembapan: 68.8%. Respon: {"message":"Data saved"}
[5/10] Terkirim -> Suhu: 30.7°C, Kelembapan: 55.7%. Respon: {"message":"Data saved"}
[6/10] Terkirim -> Suhu: 30.9°C, Kelembapan: 63.5%. Respon: {"message":"Data saved"}
[7/10] Terkirim -> Suhu: 28.5°C, Kelembapan: 60.5%. Respon: {"message":"Data saved"}
[8/10] Terkirim -> Suhu: 29.9°C, Kelembapan: 69.6%. Respon: {"message":"Data saved"}
[9/10] Terkirim -> Suhu: 30.1°C, Kelembapan: 59.4%. Respon: {"message":"Data saved"}
[10/10] Terkirim -> Suhu: 25.1°C, Kelembapan: 64.3%. Respon: {"message":"Data saved"}

=== Pengiriman Data Selesai! ===
Silakan cek halaman web Django Anda di:
-> https://m70204.belajarhub.id/django/device/device01/
-> https://m70204.belajarhub.id/django/device/device01/gauge/
```

Executed the custom Python automation script 'send_django_mock_data.py' locally. The script successfully formulated JSON payloads for temperature and humidity, sent 10 HTTP POST requests to the Django REST API endpoint at 'https://m70204.belajarhub.id/django/api/sensor/', and received success status responses from the server.

Step 3: Tabular Sensor Log Verification (device_data_view)

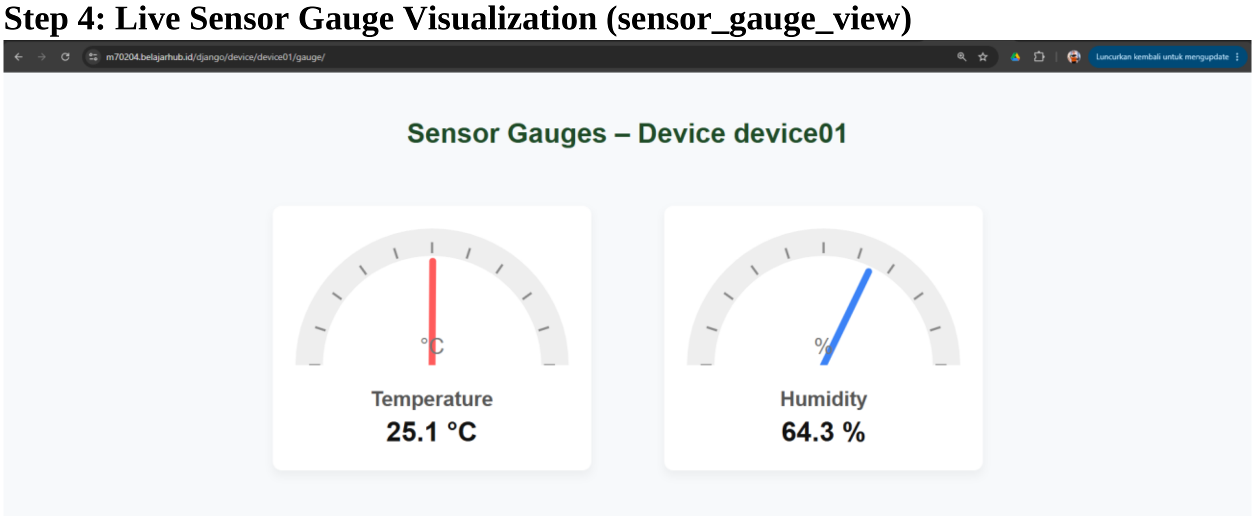


Sensor Data for device01

Timestamp	Temperature (°C)	Humidity (%)
2026-06-16 11:45:15	25.1	64.3
2026-06-16 11:45:13	30.1	59.4
2026-06-16 11:45:12	29.9	69.6
2026-06-16 11:45:11	28.5	60.5
2026-06-16 11:45:10	30.9	63.5
2026-06-16 11:45:09	30.7	55.7
2026-06-16 11:45:07	31.1	68.8
2026-06-16 11:45:06	26.3	68.1
2026-06-16 11:45:05	31.2	58.8
2026-06-16 11:45:04	29.2	60.4

Accessed the Django web application at 'https://m70204.belajarhub.id/django/device/device01/'. The page successfully queried the SQLite database, fetched the 10 recently sent sensor readings, and displayed them in an HTML table containing the correct timestamps, temperatures, and humidity percentages.

Step 4: Live Sensor Gauge Visualization (sensor_gauge_view)



Navigated to the visual gauge display at 'https://m70204.belajarhub.id/django/device/device01/gauge/'. The interface displays two semi-circular gauges representing the latest temperature (25.1 °C) and humidity (64.3 %) telemetry readings, providing real-time visual instrumentation.

<table>
<tr><th>Nilai</th><th>Tanda Tangan Dosen Pengampu / Tutor</th><th>Tanda Tangan Mahasiswa</th></tr>
<tr><td></td><td>(Prof. Jong-Dae Park)</td><td>(Muhammad Ketsar Ali Abi Wahid)</td></tr>
<tr><td colspan="2">Diserahkan pada Tanggal:</td><td>Tanggal Mengumpulkan:</td></tr>
<tr><td colspan="2">12/06/2026</td><td>16/06/2026</td></tr>
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